

EMI 2023 International Conference

Palermo, Italy, August 27 - 30, 2023

Title: Numerical Modelling and Experimental Analysis on a Scaled Unreinforced Masonry Building

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Keywords: masonry building, dynamic identification, finite element modelling, shaking table tests

Abstract

Unreinforced masonry structures represent a significant part of the European building heritage and include historic constructions with architectural and cultural value. However, recent earthquakes in Italy have dramatically demonstrated that the seismic behaviour of unreinforced masonry buildings is typically characterized by premature brittle collapse mechanisms that can cause serious consequences for the protection of human lives and for the preservation of historical and cultural heritage. Therefore, the issue related to seismic assessment and retrofit of existing masonry buildings is currently attracting considerable attention due to the progressive relative reduction of new construction activity with respect to interventions on existing structures. This highlights the necessity to improve the knowledge about the dynamic behaviour of masonry structures excited by strong ground motions, to develop useful design solutions to mitigate the seismic vulnerability of the historical constructions and of masonry buildings in general. In this regard, numerical simulation is still a very complex task due to the different construction characteristics, properties and heterogeneity of materials. Finite element modelling can be used to predict and numerically analyse the structural behaviour of masonry buildings under seismic loading [1]. The main purpose of this study is the evaluation of the seismic performance of the 2:3 scale prototype of a two-storey masonry building, built with the typical materials and construction techniques used in old masonry buildings in southern Italy. The dynamic characterization of the unreinforced masonry model was estimated from the acceleration time histories recorded during Ambient Vibration (AVT) and Impact Hammer (HT) tests. Using a macro-modelling approach, the estimation of the dynamic modal characteristics of the system has allowed the calibration of finite element (FE) numerical models, useful for simulating the dynamic response of the masonry prototype through the execution of seismic analyses, with the reproduction of different natural accelerograms of historical earthquakes, in all the three directions and with increasing intensity [2]. The numerical results obtained will support the design and execution of shaking table tests which will be performed at the LEDA Research Institute (Enna, Italy [3]), before and after seismic reinforcement works with innovative techniques.

References

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